

Grade 11 Life Sciences: Immunity Makerspace Learning Module

Comprehensive Teaching and Learning Materials

Designed for a Technology-Enhanced Learning Environment

Immune System Makerspace

Thembisile Buthelezi

Context

Target Population

- Grade 11 Life Sciences learners

Subject Area

- Human Immune System
- Focus: immune response, pathogens, antibodies

Tools and Materials

1. Digital tools

- AI-supported learning*
NotebookLM, Gens, Studyfetch
- Games*
Immune Defense Game
BioMan Biology
- Simulations*
PhET Interactive Simulations

2. Physical materials

- Cardboard, markers, glue
- Beads/string (for modeling antibodies and pathogens)
- Printed diagrams
- Microscopes for viewing cells such as blood cells

Immune System Learning Lab



AI technologies were included in this makerspace to support personalized learning and enhance learners' understanding of complex immune system processes. Tools such as NotebookLM and Gens allow learners to generate explanations, ask questions, and access additional support based on their individual needs. However, learners are also encouraged to critically evaluate AI-generated responses and compare them with scientific knowledge, ensuring that AI is used as a support tool rather than a replacement for learning.

This makerspace design aligns with Gravel and Puckett (2023), who highlight the importance of intentional design and teacher decision-making in implementing effective STEM makerspaces.

Gravel, B. E., & Puckett, C. (2023). What shapes implementation of a school-based makerspace? *International Journal of STEM Education*, 10(1), 7–22.

Learning Theories

Cognitive Load Theory

- supports this makerspace by ensuring that complex immune system concepts are broken into smaller, manageable parts across different stations. The use of visuals, simulations, and models helps reduce cognitive overload and makes learning more accessible.

Experiential Learning Theory

- Experiential Learning Theory supports this makerspace by emphasizing learning through active engagement. Learners interact with simulations, games, and hands-on activities, allowing them to experience and apply their understanding of immune system processes.

Activities

Station 1: Explore Zone

- View diagrams, posters, and short videos
- Activate prior knowledge

Station 2: Game & Simulation

- Play an immune defense game
- Interact with simulations
- Observe cause-and-effect relationships in immune responses

Station 3: Build Zone

- Construct a physical model of the immune response
- Represent interactions between pathogens and antibodies

Station 4: Apply Zone

- Analyze real-world scenarios (infections, vaccination)
- Explain how the immune system responds

Station 5: AI & Reflection Zone

- Use AI tools to generate explanations
- Compare AI responses with learned content

AI Integration

NotebookLM, Gens and Studyfetch

Table of Contents

TEACHER NOTES AND FACILITATION GUIDE	3
SUGGESTED DIGITAL TOOLS AND SIMULATIONS.....	4
EXTENSION ACTIVITIES.....	4
RESEARCH TASK.....	4
CREATIVE PROJECT	5
CASE STUDIES AND PROBLEM-BASED TASKS.....	5
CASE STUDY 1: A MYSTERIOUS ILLNESS SPREADS THROUGH THE SCHOOL	5
CASE STUDY 2: THE SOCCER INJURY THAT BECAME INFECTED	6
CASE STUDY 3: THE COMMUNITY VACCINATION CRISIS.....	7
CASE STUDY 4: THE RISE OF ANTIBIOTIC RESISTANCE.....	8
CASE STUDY 5: LIVING WITH HIV	9
STATION 1 LESSON PLAN	10
STATION 2 LESSON PLAN	12
STATION 3 LESSON PLAN	14

STATION 4 LESSON PLAN	17
STATION 5 LESSON PLAN	19
REFERENCES.....	22

Teacher Notes and Facilitation Guide

Teacher Role

The teacher acts as a facilitator, guide, questioner, and supporter throughout the learning process. Rather than relying solely on direct instruction, the teacher encourages learners to actively engage with activities, discussions, simulations, and inquiry-based tasks. The teacher supports learners by guiding exploration, asking thought-provoking questions, addressing misconceptions, and providing scaffolding where necessary to deepen understanding. In addition, the teacher promotes collaboration by encouraging group discussions, shared problem-solving, and peer interaction as learners work through the different stations and activities within the module.

Common Misconceptions

1. Antibiotics kill viruses.
 2. Vaccines cause disease.
 3. Immunity is always permanent.
 4. Inflammation is harmful only.
-

Suggested Digital Tools and Simulations

Tool	Purpose
PhET	Interactive simulations
BioMan Biology	Biology games and quizzes
HHMI BioInteractive	Scientific animations
YouTube animations	Visual explanations
NotebookLM	Content transformation
Canva	Infographics and posters

Extension Activities

Research Task

Learners research:

- COVID-19 vaccines
- HIV/AIDS
- Tuberculosis
- Antibiotic resistance

Creative Project

Learners create:

- Awareness campaigns
 - Educational videos
 - Podcasts
 - Public health posters
-

Case Studies and Problem-Based Tasks

Case Study 1: A Mysterious Illness Spreads Through the School

During the middle of winter, several learners at Siyakhula High School begin complaining of sore throats, coughing, headaches, fever, and extreme fatigue. At first, only a few learners are absent, but within one week almost half of one Grade 11 class has become sick. Teachers notice that many learners are coughing without covering their mouths, sharing water bottles, and sitting very close together during lessons.

One learner, Ayanda, becomes seriously ill and develops difficulty breathing. Another learner, however, recovers within only a few days even though both learners were exposed to the same illness at the same time. The school principal becomes concerned about how quickly the illness is spreading and invites healthcare workers to educate learners about disease prevention and the immune system.

Inquiry Questions

1. What type of pathogen is most likely responsible for this outbreak?
 2. How do respiratory diseases spread between people?
 3. Why might Ayanda have become more seriously ill than other learners?
 4. How would the immune system respond to this infection?
 5. What measures could the school introduce to reduce the spread of disease?
 6. Explain how both innate and adaptive immunity may help protect the body during this outbreak.
-

Case Study 2: The Soccer Injury That Became Infected

During an interschool soccer tournament, a learner named Thabo falls on the field and cuts his knee on rough gravel. At first, the injury seems minor, and he washes it quickly with water before continuing to play. Two days later, however, the area around the wound becomes swollen, red, warm, and painful. Thabo also notices yellow pus forming around the cut and begins to develop a fever.

His mother takes him to a clinic where the doctor explains that bacteria entered the body through the broken skin. The doctor prescribes antibiotics and explains that the body has already started fighting the infection through inflammation and immune responses. Thabo is advised to rest and complete the full course of antibiotics even if he starts feeling better after a few days.

Inquiry Questions

1. Why did the infection develop after the injury?

2. Why is the skin considered an important defense barrier?
 3. Explain why the injured area became swollen, red, and painful.
 4. What role do white blood cells play during infection?
 5. Why did the doctor prescribe antibiotics?
 6. Why is it important for Thabo to complete the full course of antibiotics?
 7. How would the body respond if the infection was left untreated?
-

Case Study 3: The Community Vaccination Crisis

A small community has recently experienced a decline in childhood vaccination rates because many parents have become concerned about misinformation shared on social media. Some parents believe vaccines are unsafe, while others think vaccines are unnecessary because many diseases are now uncommon.

Several months later, a measles outbreak occurs in the area. Many young children become sick, and local clinics quickly become overwhelmed. Healthcare workers discover that most infected children had not received the measles vaccine. However, a few vaccinated children are exposed to the disease but do not become seriously ill. Doctors and public health officials begin visiting schools and community centers to educate people about immunity, vaccination, and disease prevention. The outbreak raises important questions about public health responsibility and how communities protect vulnerable individuals.

Inquiry Questions

1. How do vaccines help protect the body against disease?

2. What are memory cells and why are they important?
 3. Why were vaccinated children less severely affected?
 4. What is herd immunity and why is it important?
 5. How can misinformation affect public health?
 6. Why are vaccines important not only for individuals but also for communities?
 7. Research how measles affects the body and explain why outbreaks can become dangerous.
-

Case Study 4: The Rise of Antibiotic Resistance

Lerato visits a clinic with a bacterial throat infection and receives antibiotics from the doctor. After taking the medication for only three days, she starts feeling much better and decides to stop taking the antibiotics because she believes she has recovered completely. A few weeks later, the infection returns, but this time the antibiotics no longer seem to work effectively.

At the same time, doctors at the local hospital report increasing numbers of patients with infections that are becoming difficult to treat. Scientists explain that some bacteria are developing resistance to antibiotics due to misuse and overuse of medication. Healthcare professionals become concerned that common infections may become more dangerous in the future if antibiotic resistance continues to increase.

Inquiry Questions

1. Why did Lerato's infection return after she stopped taking antibiotics?
2. What is antibiotic resistance?

3. How does misuse of antibiotics contribute to resistant bacteria?
 4. Why can resistant bacteria become a major public health problem?
 5. Explain why antibiotics cannot be used to treat viral infections.
 6. Research examples of antibiotic-resistant bacteria and describe their impact on healthcare systems.
 7. Suggest ways communities can reduce antibiotic resistance.
-

Case Study 5: Living with HIV

Sipho, a 22-year-old university student, is diagnosed with HIV after visiting a clinic for repeated illnesses and infections. Over time, Sipho notices that he becomes sick more frequently than before and takes longer to recover from common infections such as flu and skin infections. His doctor explains that HIV attacks helper T cells, which play an important role in coordinating immune responses.

Sipho begins antiretroviral treatment (ART) and attends counseling sessions to learn how to manage the condition. During one of the sessions, healthcare workers explain that although HIV weakens the immune system, proper treatment can help people live long and healthy lives. They also discuss ways of preventing HIV transmission and reducing stigma within communities.

Inquiry Questions

1. What type of pathogen causes HIV?
2. Which cells of the immune system are attacked by HIV?
3. How does HIV weaken the immune system?
4. Why does Sipho become more vulnerable to infections?

5. What is the purpose of antiretroviral treatment (ART)?
6. Research how HIV is transmitted and how transmission can be prevented.
7. Why is education about HIV/AIDS important for public health and reducing stigma?

Station 1 Lesson Plan

Section	Content
Station	Explore Zone
Topic	Introduction to the Immune System and Pathogens
Grade	Grade 11 Life Sciences
Duration	45–60 Minutes
Station Purpose	This station introduces learners to the immune system by activating prior knowledge and helping learners explore the relationship between pathogens, disease, and body defense mechanisms through visuals, diagrams, short videos, and guided discussion.
Learning Objectives	• Define immunity • Define a pathogen • Identify different types of pathogens • Explain why the immune system is important • Recognize that the body has defense mechanisms against disease
Resources Needed	Digital resources: short videos, immune system diagrams, pathogen posters, interactive slides. Physical resources: worksheets, sticky notes, printed diagrams, pens/highlighters.
Key Vocabulary	Immunity, pathogen, immune system, infection, virus, bacteria
Introduction Activity	The teacher begins with questions such as: “Why do people get sick?” and “How does the body protect itself?” Learners discuss prior experiences with illness and brainstorm ideas about immunity.

Teacher Activity: Introduction	The teacher facilitates discussion, introduces the term “immunity,” and explains that the body has defense systems against harmful microorganisms.
Learner Activity: Introduction	Learners share experiences of illness, discuss diseases they know, and write one idea about immunity on sticky notes.
Main Exploration Activity	Learners rotate through posters, diagrams, videos, and visuals showing pathogens and the immune system.
Teacher Activity: Main Lesson	The teacher explains: • What pathogens are • Types of pathogens (bacteria, viruses, fungi, parasites) • The role of the immune system • How pathogens cause disease
Learner Activity: Main Lesson	Learners observe diagrams, complete worksheets, identify pathogen types, and discuss observations in groups.
Video Reflection Activity	Learners watch a short immunity video or animation and answer guided reflection questions.
Teacher Activity: Video Reflection	The teacher pauses the video to explain important concepts and facilitate discussion.
Learner Activity: Video Reflection	Learners discuss what surprised them, why the immune system is important, and what might happen without immunity.
Interactive Group Task	Learners work with pathogen cards/images containing diseases, symptoms, and pathogen types.
Teacher Activity: Group Task	The teacher distributes pathogen cards and guides learners through the activity.
Learner Activity: Group Task	Learners identify pathogen types, discuss disease spread, and share findings with the class.

Conclusion	The teacher summarizes that the body is constantly exposed to pathogens and that the immune system protects the body from disease.
Informal Assessment	Observation, questioning, worksheet completion, group participation, discussion responses
Exit Ticket	1. Define immunity. 2. What is a pathogen? 3. Name one type of pathogen.
Homework / Extension Activity	Learners research one viral disease and one bacterial disease, including symptoms and how they spread.

Station 2 Lesson Plan

Section	Content
Station	Game & Simulation Zone
Topic	Innate Immunity and the Body's First Line of Defense
Grade	Grade 11 Life Sciences
Duration	45–60 Minutes
Station Purpose	This station allows learners to explore how the body responds immediately when pathogens enter the body. Through simulations, games, and interactive activities, learners investigate innate immunity, physical barriers, inflammation, and the role of white blood cells in defending the body.
Learning Objectives	• Define innate immunity • Explain the body's first line of defense • Describe how pathogens enter the body • Explain the role of white blood cells • Describe inflammation as an immune response • Observe cause-and-effect relationships in immune responses
Resources Needed	Digital resources: immune defense game, immune response simulations, animations, interactive diagrams. Physical resources: worksheets, pens/highlighters, headphones (optional).

Key Vocabulary	Innate immunity, inflammation, white blood cells, phagocyte, pathogen, barrier defense
Introduction Activity	The teacher revises concepts from Station 1 by asking learners how pathogens enter the body and what happens when germs enter through cuts or wounds. Learners predict how the body responds to infection.
Teacher Activity: Introduction	The teacher introduces innate immunity as the body's immediate and non-specific defense system. The teacher explains that learners will use simulations and games to observe immune responses occurring inside the body.
Learner Activity: Introduction	Learners discuss prior experiences with cuts, swelling, fever, or infection and share predictions about how the body responds to pathogens.
Main Simulation Activity	Learners interact with simulations and games showing how pathogens invade the body and how immune cells respond.
Teacher Activity: Main Lesson	The teacher explains: • The first line of defense (skin, mucus, tears, stomach acid) • How pathogens enter the body • The role of white blood cells • Phagocytosis • Inflammation and healing processes
Learner Activity: Main Lesson	Learners interact with simulations, observe immune responses, complete guided worksheets, and identify the stages of innate immune responses.
Guided Observation Task	Learners focus on cause-and-effect relationships observed in the simulations.
Teacher Activity: Guided Observation	The teacher pauses simulations to explain movement of white blood cells, inflammation, and destruction of pathogens.
Learner Activity: Guided Observation	Learners record observations about how the body responds during infection and create a simple immune response flow diagram.
Interactive Group Discussion	Learners discuss why inflammation occurs and why fever can help the body fight infection.

Teacher Activity: Group Discussion	The teacher facilitates discussion using questions such as: “Why do cuts become swollen?” and “What might happen without innate immunity?”
Learner Activity: Group Discussion	Learners discuss observations, compare simulation outcomes, and explain immune responses using scientific vocabulary.
Conclusion	The teacher summarizes that innate immunity is fast, immediate, and protects the body from pathogens using barriers and white blood cells.
Informal Assessment	Observation, questioning, simulation participation, worksheet completion, group discussion
Exit Ticket	1. What is innate immunity? 2. Name one barrier defense. 3. What is the role of white blood cells?
Homework / Extension Activity	Learners write a short paragraph explaining how the body responds when pathogens enter through a cut in the skin.

Station 3 Lesson Plan

Section	Content
Station	Build Zone
Topic	Adaptive Immunity, Antibodies, and Immune Memory
Grade	Grade 11 Life Sciences
Duration	45–60 Minutes
Station Purpose	This station helps learners understand how the body develops specific immune responses against pathogens. Learners explore adaptive immunity through model-building activities, diagrams, and collaborative tasks that demonstrate how antibodies, lymphocytes, and memory cells protect the body.

Learning Objectives	• Define adaptive immunity • Distinguish between innate and adaptive immunity • Explain the role of B cells and T cells • Describe how antibodies function • Explain immune memory • Construct models representing immune responses
Resources Needed	Physical resources: cardboard, beads, clay, pipe cleaners, markers, glue, string, scissors, printed immune system diagrams. Digital resources: animations of adaptive immunity, antibody diagrams, immune response videos.
Key Vocabulary	Adaptive immunity, antibody, antigen, lymphocyte, B cell, T cell, immune memory, memory cell
Introduction Activity	The teacher begins by revising innate immunity and asks learners why some diseases are only contracted once. Learners discuss why vaccinated individuals often do not become severely ill when exposed to certain diseases.
Teacher Activity: Introduction	The teacher introduces adaptive immunity as a specific immune response that develops after exposure to pathogens. The teacher explains that adaptive immunity remembers pathogens and responds more effectively during future infections.
Learner Activity: Introduction	Learners discuss examples of immunity, vaccinations, and diseases they may have experienced only once.
Main Learning Activity	Learners explore how antibodies and immune cells respond to pathogens through model-building and guided exploration activities.
Teacher Activity: Main Lesson	The teacher explains: • Adaptive immunity is specific • B cells produce antibodies • T cells destroy infected cells • Antibodies attach to pathogens • Memory cells remain after infection • Future immune responses become faster and stronger
Learner Activity: Main Lesson	Learners observe diagrams and videos showing adaptive immune responses and complete guided worksheet activities.

Build Activity	Learners construct physical models representing interactions between pathogens, antibodies, and immune cells.
Teacher Activity: Build Activity	The teacher demonstrates how learners can use materials such as beads, clay, and string to represent pathogens, antibodies, and immune cells. The teacher guides learners during construction.
Learner Activity: Build Activity	Learners work in groups to build models showing: • Pathogen invasion • Antibody attachment • B cell activity • T cell response • Formation of memory cells
Guided Discussion	Learners discuss how adaptive immunity differs from innate immunity and why memory cells are important.
Teacher Activity: Guided Discussion	The teacher facilitates discussion using questions such as: “Why does the body respond faster during a second infection?” and “How do vaccines use immune memory?”
Learner Activity: Guided Discussion	Learners explain adaptive immune responses using their models and scientific terminology.
Conclusion	The teacher summarizes that adaptive immunity is specific, develops after exposure to pathogens, and provides long-term protection through immune memory.
Informal Assessment	Observation, questioning, model construction, group participation, worksheet completion
Exit Ticket	1. What is adaptive immunity? 2. What is the role of antibodies? 3. Why are memory cells important?
Homework / Extension Activity	Learners create a labeled diagram showing how antibodies respond to pathogens and write a short explanation of immune memory.

Station 4 Lesson Plan

Section	Content
Station	Create Zone
Topic	Vaccination, Immune Memory, and Public Health
Grade	Grade 11 Life Sciences
Duration	45–60 Minutes
Station Purpose	This station helps learners understand how vaccines work and how vaccination contributes to individual and community protection. Learners explore immune memory, herd immunity, and disease prevention through creative activities, public health investigations, and awareness campaign design tasks.
Learning Objectives	• Explain how vaccines work • Describe immune memory • Explain herd immunity • Discuss the importance of vaccination in public health • Analyze the impact of low vaccination rates • Create awareness materials related to vaccination
Resources Needed	Physical resources: poster paper, markers, colored pencils, glue, magazines, scissors, learner worksheets. Digital resources: vaccination videos, WHO/CDC visuals, public health infographics, short animations on vaccines and immune memory.
Key Vocabulary	Vaccination, vaccine, immune memory, herd immunity, immunity, outbreak, public health
Introduction Activity	The teacher begins by asking learners whether they have received vaccines before and why vaccines are important. Learners discuss diseases they know that can be prevented through vaccination.
Teacher Activity: Introduction	The teacher introduces vaccination as a method of stimulating the immune system to produce antibodies and memory cells without causing severe disease.

Learner Activity: Introduction	Learners share ideas about vaccines, discuss prior experiences with vaccination, and brainstorm why governments encourage immunization programs.
Main Learning Activity	Learners investigate how vaccines prepare the immune system to respond quickly during future infections.
Teacher Activity: Main Lesson	The teacher explains: • Vaccines contain weakened or inactive pathogens • Vaccines stimulate antibody production • Memory cells remain after vaccination • The immune system responds faster during future infections • Herd immunity protects communities
Learner Activity: Main Lesson	Learners observe videos, diagrams, and infographics explaining vaccination and complete guided worksheet questions.
Creative Public Health Activity	Learners create vaccination awareness campaigns using posters, infographics, slogans, or digital content.
Teacher Activity: Creative Activity	The teacher provides examples of public health campaigns and guides learners in developing accurate scientific messages about vaccination.
Learner Activity: Creative Activity	Learners work in groups to create awareness materials promoting vaccination and disease prevention.
Case Study Discussion	Learners analyze a disease outbreak scenario in a community with low vaccination rates.
Teacher Activity: Case Study	The teacher presents a real-world or fictional outbreak case study and facilitates discussion about how vaccines reduce disease spread.
Learner Activity: Case Study	Learners discuss why diseases spread more rapidly in unvaccinated populations and explain how herd immunity works.

Guided Discussion	Learners discuss myths and misconceptions about vaccines and evaluate scientific evidence related to vaccination.
Teacher Activity: Guided Discussion	The teacher facilitates respectful discussion and addresses common misconceptions about vaccines.
Learner Activity: Guided Discussion	Learners ask questions, share opinions, and support responses using scientific reasoning.
Conclusion	The teacher summarizes that vaccines help the immune system develop memory, reduce disease spread, and protect both individuals and communities.
Informal Assessment	Observation, participation, group discussion, worksheet completion, creative project participation
Exit Ticket	1. How do vaccines work? 2. What is herd immunity? 3. Why are memory cells important?
Homework / Extension Activity	Learners research one vaccine-preventable disease and prepare a short summary explaining how vaccination helps control the disease.

Station 5 Lesson Plan

Section	Content
Station	Investigate Zone
Topic	HIV/AIDS, Antibiotics, and Antibiotic Resistance
Grade	Grade 11 Life Sciences
Duration	45–60 Minutes

Station Purpose	This station helps learners investigate how diseases and medicines affect the immune system. Learners explore HIV/AIDS, the use of antibiotics, and the growing problem of antibiotic resistance through inquiry-based learning, case studies, and problem-solving activities.
Learning Objectives	• Explain how HIV affects the immune system • Describe the role of helper T cells • Explain how antibiotics work • Distinguish between bacterial and viral infections • Explain antibiotic resistance • Analyze the consequences of antibiotic misuse
Resources Needed	Digital resources: HIV animations, videos on antibiotic resistance, interactive diagrams, case study slides. Physical resources: worksheets, printed case studies, pens/highlighters, chart paper.
Key Vocabulary	HIV, AIDS, helper T cell, antibiotic, antibiotic resistance, bacteria, virus, immune deficiency
Introduction Activity	The teacher begins by asking learners: “Why do some diseases weaken the immune system?” and “Why can’t antibiotics cure flu viruses?” Learners discuss prior knowledge about HIV/AIDS and antibiotics.
Teacher Activity: Introduction	The teacher introduces HIV as a virus that attacks helper T cells and explains that antibiotics are medicines used to treat bacterial infections.
Learner Activity: Introduction	Learners share ideas about HIV/AIDS, antibiotics, and disease treatment and predict what happens when the immune system becomes weak.
Main Learning Activity	Learners investigate how HIV affects the immune system and how misuse of antibiotics contributes to antibiotic resistance.
Teacher Activity: Main Lesson	The teacher explains: • HIV attacks helper T cells • A weakened immune system struggles to fight infection • AIDS develops when the immune system becomes severely damaged • Antibiotics kill bacteria, not viruses • Overuse and misuse of antibiotics lead to resistant bacteria
Learner Activity: Main Lesson	Learners observe diagrams and videos explaining HIV infection and antibiotic resistance and complete guided worksheet activities.

Investigation Activity	Learners analyze case studies involving antibiotic misuse and spread of infectious disease.
Teacher Activity: Investigation Activity	The teacher presents real-world or fictional case studies involving incomplete antibiotic treatment, misuse of medication, or spread of HIV.
Learner Activity: Investigation Activity	Learners work in groups to analyze what went wrong in each case study and propose solutions to reduce disease spread and antibiotic resistance.
Problem-Solving Discussion	Learners discuss why antibiotic resistance is a growing global health problem.
Teacher Activity: Discussion	The teacher facilitates discussion using questions such as: “Why should antibiotics be completed?” and “Why is antibiotic resistance dangerous?”
Learner Activity: Discussion	Learners explain how resistant bacteria survive and spread and discuss ways to use antibiotics responsibly.
Conclusion	The teacher summarizes that HIV weakens the immune system, antibiotics only treat bacterial infections, and misuse of antibiotics contributes to antibiotic resistance.
Informal Assessment	Observation, group participation, worksheet completion, case study analysis, questioning
Exit Ticket	1. How does HIV affect the immune system? 2. Why can’t antibiotics treat viral infections? 3. What is antibiotic resistance?
Homework / Extension Activity	Learners research one example of antibiotic-resistant bacteria and explain why antibiotic resistance is a major public health concern.

References

Black, P., & Wiliam, D. (1998). Assessment and classroom learning.

Bransford, J. D., Brown, A. L., & Cocking, R. R. (2000). How People Learn.

Department of Basic Education. (2011). CAPS Life Sciences Grades 10–12.

Kolb, D. A. (1984). Experiential Learning.

Prince, M. (2004). Does active learning work?

Rutten, N., van Joolingen, W. R., & van der Veen, J. T. (2012). Simulations in science education.

Sweller, J., Ayres, P., & Kalyuga, S. (2011). Cognitive Load Theory.